

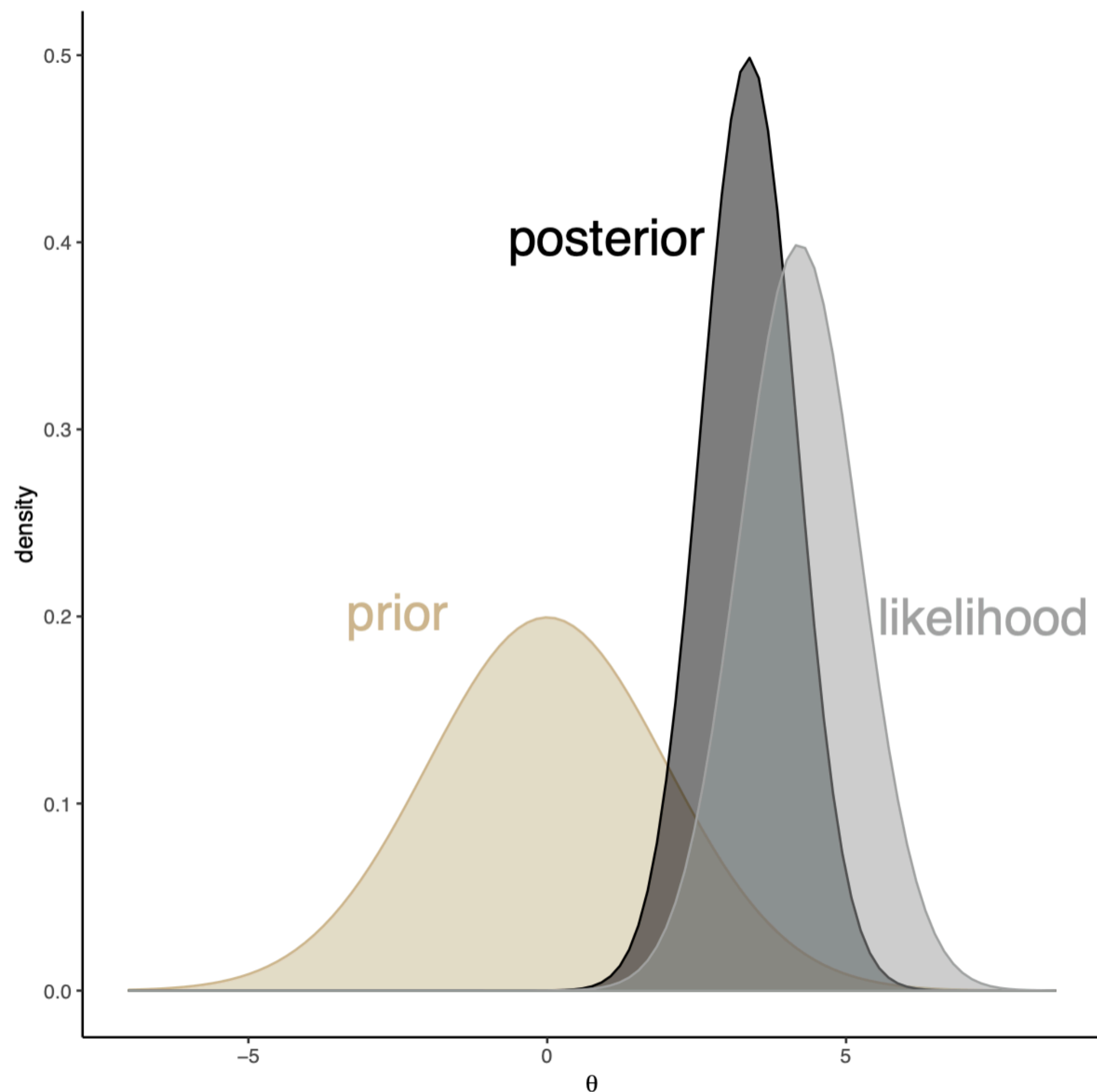
Unit #3: Bayesian inference on multiple parameters

Bayesian Statistical Methods, 2.1.6, 2.1.7, 4.1.2, 4.2.1,
4.2.2

Secondary reference: Bayesian Data Analysis, 3.1 - 3.3

Photo by [Alexis Gethin](#) on [Unsplash](#)



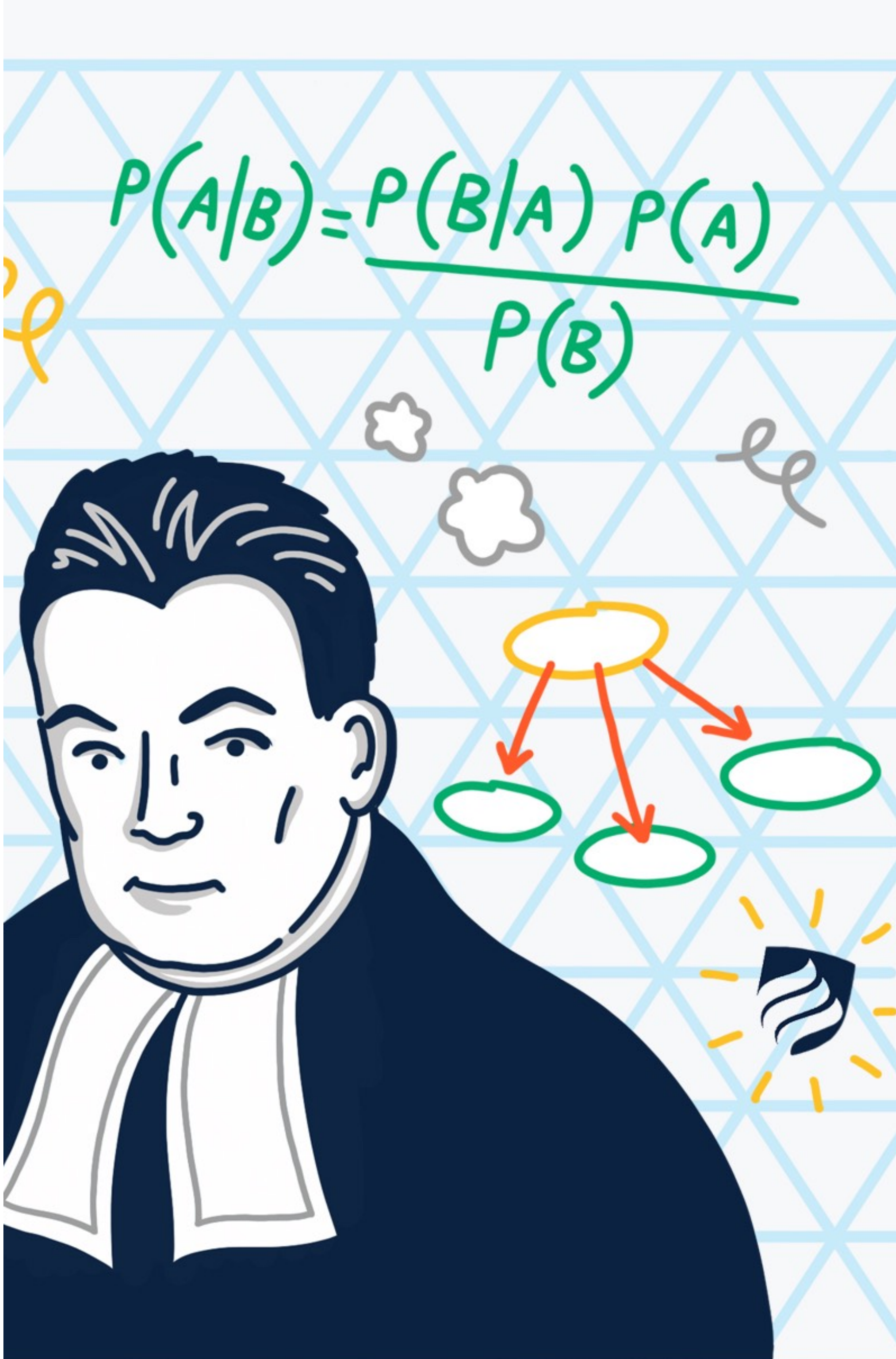


So far, we've mainly looked at single-parameter Bayesian inference.

But often, we will be interested in problems where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)^T$.

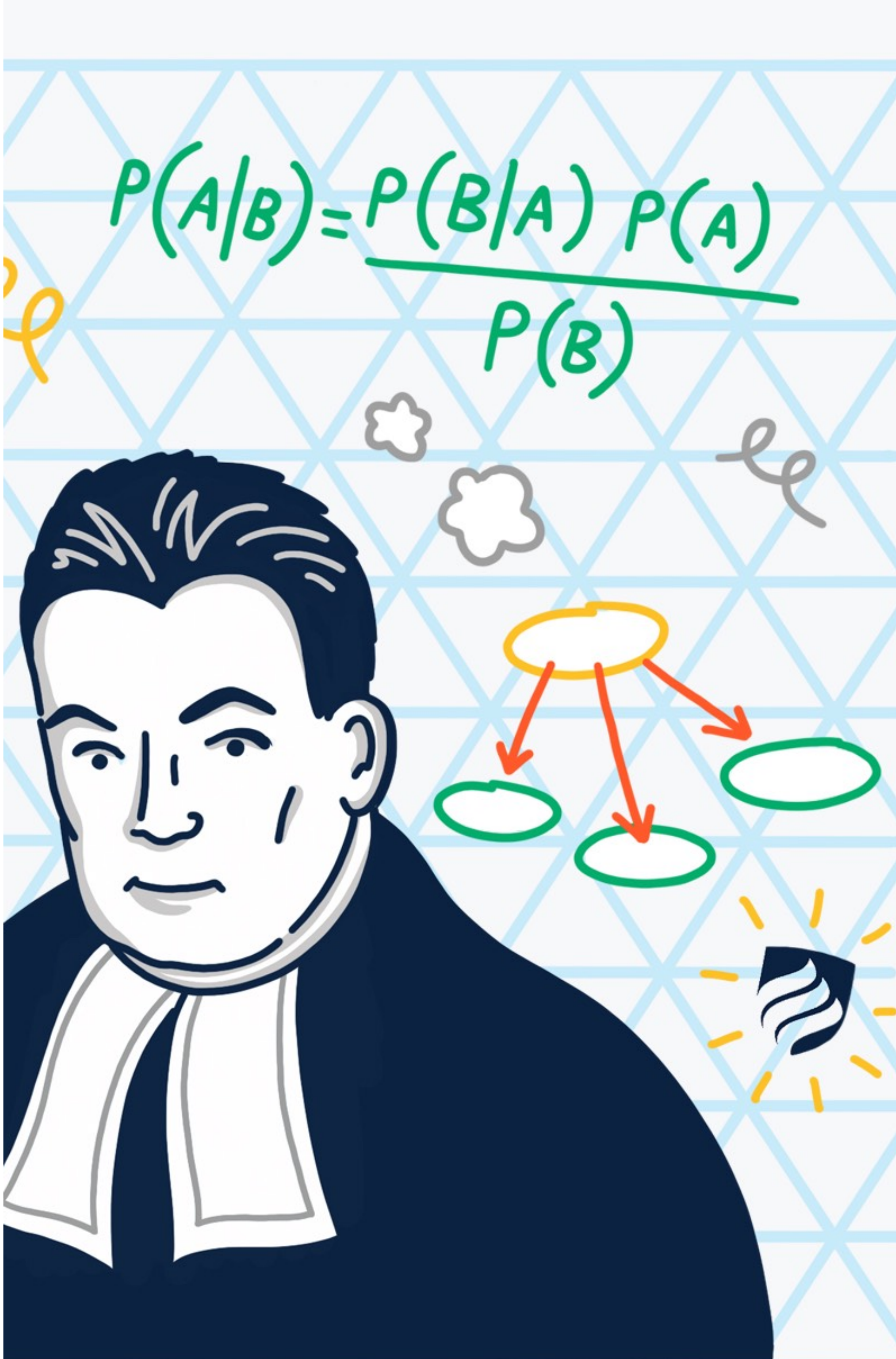
In many problems inference is only required on a small number of the unknown parameters, even though other parameters are present in the model. Parameters of this kind are often called **nuisance parameters**.

Example: Often, we have the model $X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where both μ, σ^2 are unknown, but we are only interested in inferences about μ .



Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(\mathbf{x} | \boldsymbol{\theta})$ where $\boldsymbol{\theta} = (\theta_1, \theta_2)^T$. Let $\pi(\boldsymbol{\theta})$ be the joint prior on the parameter vector $\boldsymbol{\theta}$. Then:

If we are mainly concerned with inferences on θ_1 , and θ_2 is a nuisance parameter, then we can find the marginal posterior for θ_1 given the data:



Posterior distributions can be computed by marginal and conditional simulation, first drawing θ_2 from its marginal posterior distribution and then θ_1 from its conditional posterior distribution, given the drawn value of θ_2 .



Photo by Toby Elliott on [Unsplash](#)

Example: Let $X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Let $\pi_\mu(\mu) = 1$ and $\pi_{\sigma^2}(\sigma^2) = 1 / \sigma^2$.

Find the posterior distribution of $\mu, \sigma^2 \mid \mathbf{x}$.

- Prior:
- Likelihood:
- Posterior:



Photo by Toby Elliott on [Unsplash](#)

The conditional posterior distribution, $\pi(\mu \mid \sigma^2, \mathbf{x})$:

The marginal posterior distribution, $\pi(\sigma^2 \mid \mathbf{x})$:



Modeling the variance:

The **inverse- χ^2 distribution** with $\nu_0 > 0$ degrees of freedom is equivalent to the inverse- Γ distribution with $\alpha = \frac{\nu_0}{2}$ and $\beta = \frac{1}{2}$.

The **scaled-inverse- χ^2 distribution** is useful for σ^2 in a normal model. The scaled-inverse- χ^2 distribution is equivalent to the inverse- Γ distribution with $\alpha = \frac{\nu_0}{2}$ and $\beta = \frac{\nu_0 \sigma_0^2}{2}$.

For simulation purposes: to obtain a random draw from the scaled-inverse- χ^2 distribution, i.e., $\theta \sim \text{scaled-inverse-}\chi^2(\nu_0, \sigma_0^2)$:

- Draw $\gamma \sim \chi^2(\nu_0) \equiv \Gamma\left(\frac{\nu_0}{2}, \frac{1}{2}\right)$
- Let $\theta = \nu_0 \sigma_0^2 / \gamma$.



Normal data with a conjugate prior distribution

Likelihood/data: $X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$

Priors:

1. $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2 / k_0)$

2. $\sigma^2 \sim \text{inv-}\Gamma\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right) \equiv \text{Scale-Inv-}\chi^2(\nu_0, \sigma_0^2)$

3. $(\mu, \sigma^2) \sim \text{normal-inverse-}\Gamma\left(\mu, \sigma^2 \mid \mu_0, \sigma_0^2 / k_0; \nu_0, \sigma_0^2\right)$

$\pi(\mu, \sigma^2) =$



Photo by Toby Elliott on Unsplash

Normal data with a conjugate prior distribution

Likelihood/data: $X_1, \dots, X_n \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$

Posterior:

$$\mu, \sigma^2 \mid \mathbf{x} \sim \text{normal-inverse-}\Gamma \left(\mu, \sigma^2 \mid \mu_n, \sigma_n^2 / k_n; v_n, \sigma_n^2 \right),$$

$$k_n = k_0 + n$$

$$\mu_n = \frac{k_0}{k_n} \mu_0 + \frac{n}{k_n} \bar{x}$$

$$v_n = v_0 + n$$

$$\sigma_n^2 = \frac{v_0 \sigma_0^2 + (n - 1) S^2 + \frac{k_0 n}{k_n} (\bar{x} - \mu_0)^2}{v_n}$$

Proof for HW!

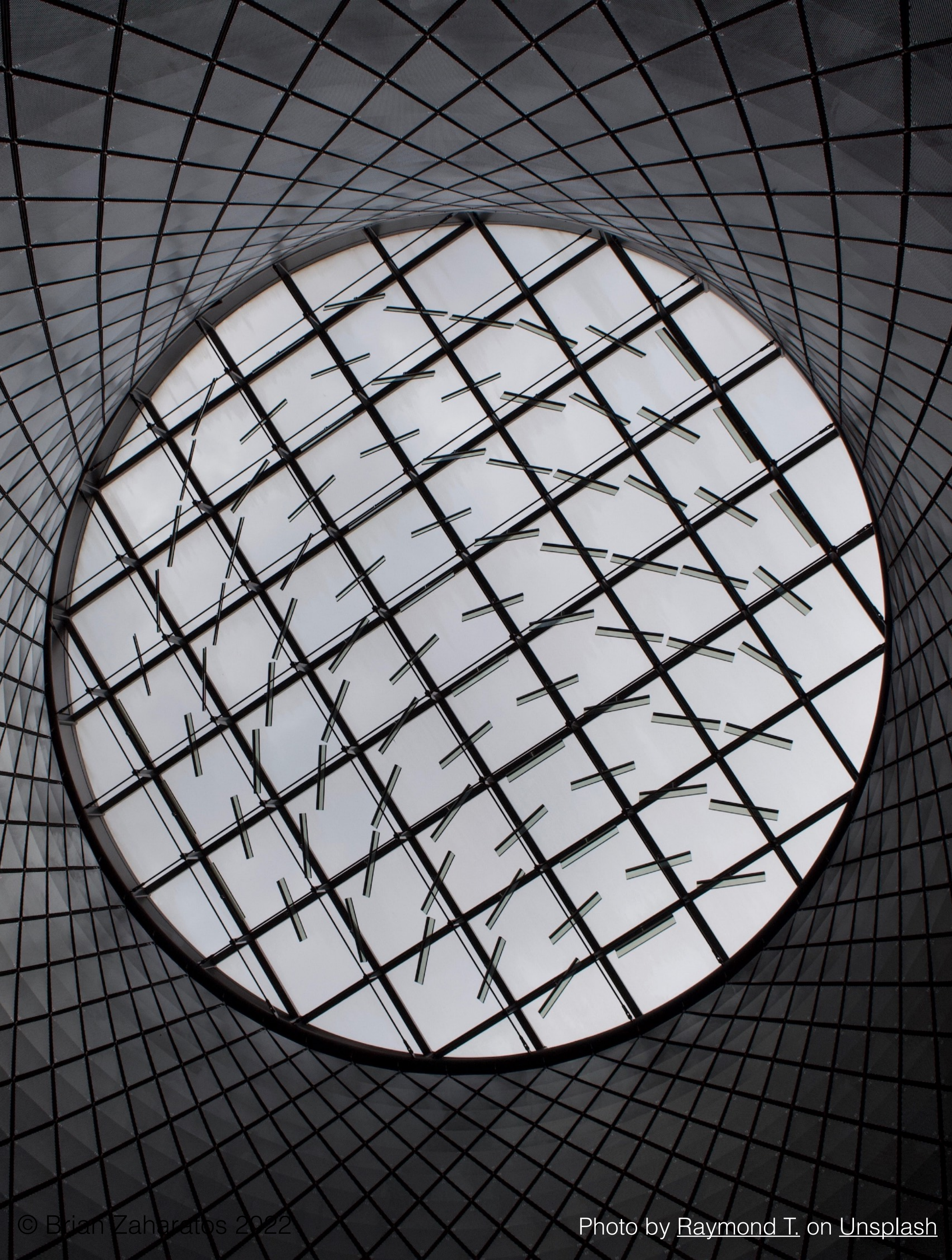
Suppose that $\mathbf{Y} \mid \boldsymbol{\beta} \sim N(X\boldsymbol{\beta}, \Sigma_n)$, where X is an $n \times p$ design matrix, $\boldsymbol{\beta}$ is a $p \times 1$ vector of parameters, and Σ_n is a variance-covariance matrix. Let $\boldsymbol{\beta} \sim N(\boldsymbol{\mu}, \Sigma_p)$. Find $\pi(\boldsymbol{\beta} \mid \mathbf{Y})$.

- Prior:

- Likelihood:

Suppose that $\mathbf{Y} \mid \boldsymbol{\beta} \sim N(X\boldsymbol{\beta}, \Sigma_n)$, where X is an $n \times p$ design matrix, $\boldsymbol{\beta}$ is a $p \times 1$ vector of parameters, and Σ_n is a variance-covariance matrix. Let $\boldsymbol{\beta} \sim N(\boldsymbol{\mu}, \Sigma_p)$. Find $\pi(\boldsymbol{\beta} \mid \mathbf{Y})$.

- Posterior:



A review of Frequentist regression assumptions:

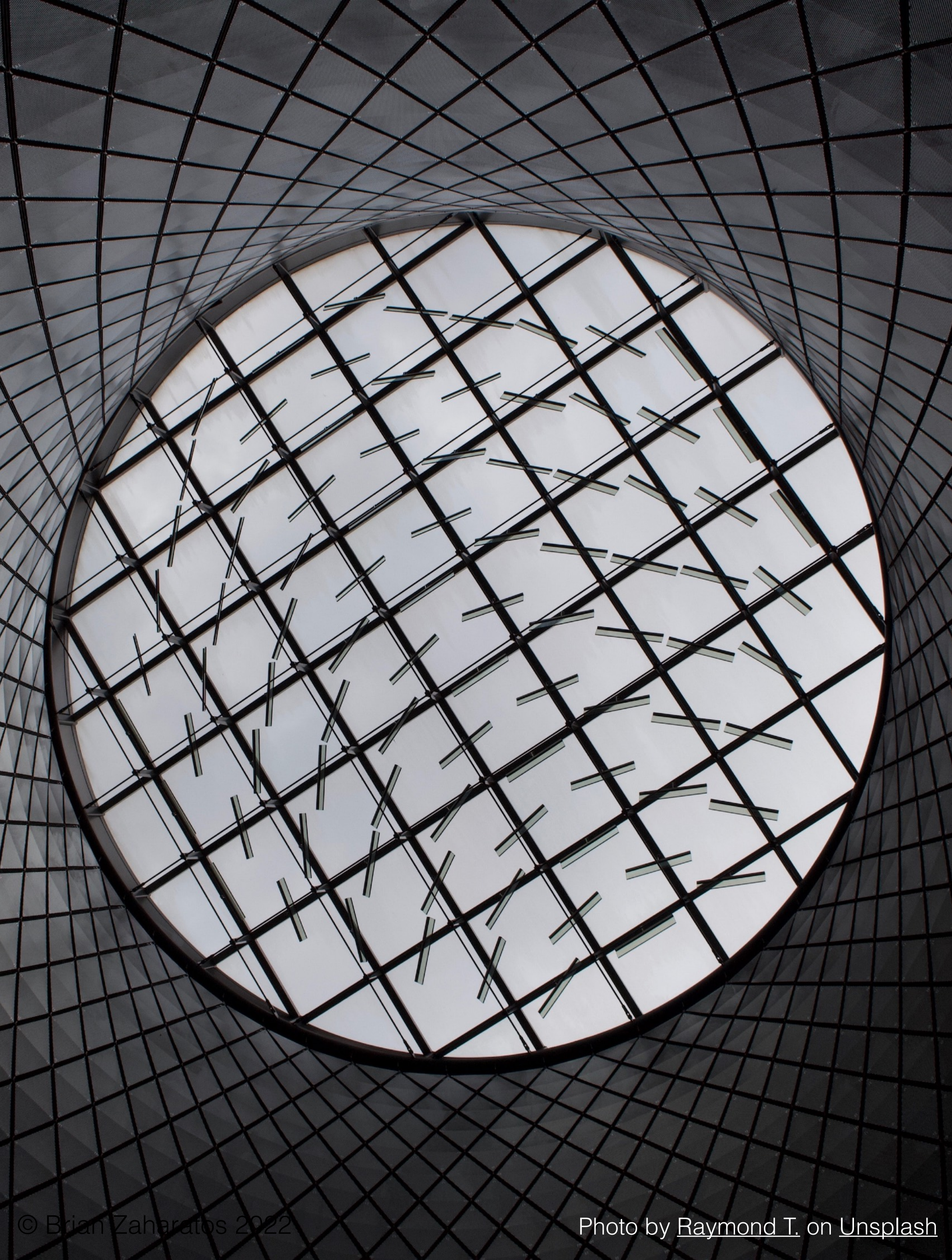
1. $E(\varepsilon_i) = 0$ for all $i = 1, \dots, n$.

2. $E(Y_i) = \mathbf{x}_i^T \boldsymbol{\beta}$ for all $i = 1, \dots, n$.

3.
$$\text{Cov}(\varepsilon_i, \varepsilon_j) = \begin{cases} 0 & i \neq j \\ \sigma^2 & i = j \end{cases}$$

4. $(X^T X)^{-1}$ exists.

5. $Y_i \stackrel{iid}{\sim} N(\mathbf{x}_i^T \boldsymbol{\beta}, \sigma^2)$



Recall that the least squares (frequentist) estimator in regression is $\hat{\boldsymbol{\beta}} = \underbrace{(X^T X)^{-1} X^T}_{\text{constant}} \mathbf{Y}$.

This estimator is a linear combination of normal random variables. Thus, it's normal with parameters:

What about the Bayes case?



Let $\mathbf{X}_i = (X_1, \dots, X_p)^T \sim N_p(\mathbf{0}, \Sigma_p)$, for $i = 1, \dots, n$. Let's consider the multivariate problem of estimating Σ_p , the variance-covariance matrix of the data.

Note that Σ_p is the same across each $\mathbf{X}_i$. Let $D = (\mathbf{X}_1, \dots, \mathbf{X}_n)$ be a random sample from $N_p(\mathbf{0}, \Sigma_p)$.



Modeling the variance-covariance matrix:

The **inverse Wishart distribution** with ν degrees of freedom and symmetric positive definite $p \times p$ scale matrix $\mathbf{R}$ is a generalization of the scaled-inverse- χ^2 distribution. It is useful for modeling Σ in a multivariate normal model.



Let $\mathbf{X}_i = (X_1, \dots, X_p)^T \sim N_p(\mathbf{0}, \Sigma_p)$, for $i = 1, \dots, n$. We'd like to estimate (i.e., find a posterior distribution for) Σ_p . Note that Σ_p is the same across each $\mathbf{X}_i$. Let $D = (\mathbf{X}_1, \dots, \mathbf{X}_n)$ be a random sample from $N_p(\mathbf{0}, \Sigma_p)$.

$M_{p \times p}$ has a **Wishart distribution** $M \sim W_p(v, R)$ where:

1. $v > p - 1$ is the degrees of freedom.
2. R is a symmetric positive definite scale matrix ($p \times p$).

The pdf of M is given by:

$$f_M(\mathbf{m}) = \left(\frac{1}{2^{vp/2} |R|^{v/2} \Gamma_p(v/2)} \right) |\mathbf{m}|^{(v-p-1)/2} \exp \left\{ -\frac{1}{2} \text{tr} (R^{-1} \mathbf{m}) \right\}$$



A prior on Σ_p

$Q = M_{p \times p}^{-1}$ has an **Inverse Wishart distribution** $Q \sim InvW_p(v, R)$ with a pdf of Q is given by:

$$f_M(\mathbf{q}) = \left(\frac{1}{2^{vp/2} |R|^{-v/2} \Gamma_p(v/2)} \right) |\mathbf{q}|^{-(v+p+1)/2} \exp \left\{ -\frac{1}{2} tr(\mathbf{q}^{-1} \mathbf{R}) \right\}$$

with $v > p - 1$. The prior concentration around

$$E(Q) = \frac{R}{v - p - 1} \text{ increases with } v.$$



Inference on a covariance matrix

Let $\mathbf{X}_i = (X_1, \dots, X_p)^T \sim N_p(\mathbf{0}, \Sigma_p)$, for $i = 1, \dots, n$. We'd like to estimate (i.e., find a posterior distribution for) Σ_p . Note that Σ_p is the same across each $\mathbf{X}_i$.

Let $D = (\mathbf{X}_1, \dots, \mathbf{X}_n)$ be a random sample from $N_p(\mathbf{0}, \Sigma_p)$. The Likelihood function is:

So, let's find $\pi(\Sigma_p \mid D)$.